

Types of Machine Learning (Based on Supervision)

In this session, we discuss the classification of Machine Learning (ML) algorithms based on the **amount of supervision** they need during training. This is the most famous way to categorize ML.

1. High-Level Categorization

Based on how much external supervision or labeling is required, we divide ML into four main branches:

1. **Supervised Machine Learning**
2. **Unsupervised Machine Learning**
3. **Semi-supervised Machine Learning**
4. **Reinforcement Learning**

2. Supervised Machine Learning

Explanation

Supervised learning is like a student learning under the guidance of a teacher. In this type, the algorithm is trained on a "Labeled Dataset." This means for every input provided to the system, the correct answer (output) is also provided.

The Goal: To find a mathematical relationship between the input (X) and the output (y) so that for a new, unseen input, the model can predict the correct output.

Example: Placement Prediction

Imagine we have data for 5,000 students.

IQ (Input)	CGPA (Input)	Placement (Output/Label)
87	7.1	Yes
111	8.9	Yes
75	6.3	No

- **Inputs:** IQ and CGPA (Features)
- **Output:** Placement status (Target)

Teacher's Reasoning: The model looks at this data, learns the patterns (e.g., "High IQ + High

CGPA usually leads to Yes"), and creates a function. When a new student arrives with IQ 90 and CGPA 8.0, the model uses its learned logic to predict "Yes" or "No."

Sub-types of Supervised Learning

To understand these, we must first know the two types of data:

1. **Numerical Data:** Numbers (Age, Weight, Salary, IQ).
2. **Categorical Data:** Groups/Labels (Gender, Nationality, Yes/No).

A. Regression

- **Definition:** When the **Output column is Numerical**, the problem is called Regression.
- **Example:** Predicting the exact salary package (e.g., 4.5 LPA, 6.2 LPA) based on IQ and CGPA.

B. Classification

- **Definition:** When the **Output column is Categorical**, the problem is called Classification.
- **Example:** Predicting if a student will be placed ("Yes") or not ("No"). Other examples include:
 - **Spam Detection:** Is this email "Spam" or "Not Spam"?
 - **Weather:** Will it "Rain" or "Not Rain" today?

3. Unsupervised Machine Learning

Explanation

In Unsupervised learning, we only have **Input data**. There is **no Output/Label**. The model is not told what the "right answer" is. Instead, it tries to find hidden patterns or structures within the data on its own.

Key Tasks in Unsupervised Learning:

I. Clustering

- **Logic:** Grouping similar data points together.
- **Example:** On a graph of IQ vs CGPA, the algorithm might find three distinct groups:
 1. High IQ, High CGPA (Top students).
 2. Low IQ, Low CGPA (Struggling students).
 3. High IQ, Low CGPA (Smart but lazy).
- **Use Case:** Customer segmentation in E-commerce (Grouping customers by buying behavior).

II. Dimensionality Reduction

- **Logic:** Reducing the number of input columns (dimensions) without losing the core information.
- **Why use it?** If you have 1,000 input columns, the model becomes very slow.

- **Feature Extraction Example:** If you have "Number of Rooms" and "Number of Washrooms," you can combine them into one column called "Square Footage" to simplify the data.
- **Visualization:** Reducing high-dimensional data (like 784 pixels in an image) down to 2D or 3D so humans can see the relationship on a graph (e.g., using t-SNE or PCA).

III. Anomaly Detection

- **Logic:** Identifying "Outliers" or unusual data points that don't fit the pattern.
- **Example:** In credit card transactions, if most transactions are \$10-\$50 but suddenly there is a \$5,000 transaction, the system flags it as a "Fraud" (Anomaly).

IV. Association Rule Learning

- **Logic:** Finding rules that describe your data, such as "People who buy X also tend to buy Y."
- **The "Beer and Diaper" Story:** A famous study (often linked to Walmart) found that men buying diapers on Fridays also tended to buy beer. By placing them together, sales increased.

4. Semi-supervised Machine Learning

Explanation

This is a middle ground. Labeling data is expensive and time-consuming because it requires humans. However, gathering raw data is cheap. Semi-supervised learning uses a **small amount of labeled data** and a **large amount of unlabeled data**.

Example: Google Photos

1. **Unsupervised Step:** Google Photos automatically groups all photos with similar faces into clusters (Clustering).
2. **Supervised Step:** You go to one photo and label it "Dad."
3. **Result:** The system automatically applies the label "Dad" to all other photos in that cluster. You only labeled *one* photo, and the system handled the rest.

5. Reinforcement Learning (RL)

Explanation

In RL, there is **no predefined dataset**. The algorithm (called an **Agent**) learns by interacting with an **Environment**. It learns through "Trial and Error."

The Loop:

1. **Action:** The Agent performs a task.
2. **Observation:** The Agent sees the result.
3. **Reward/Punishment:** * If the result is good $\rightarrow$ Positive Reward (+ points).

- If the result is bad $\rightarrow$ Punishment/Penalty (- points).
- 4. **Policy Update:** The Agent updates its "Rulebook" (Policy) to repeat actions that get rewards and avoid actions that lead to punishment.

Examples:

- **Self-Driving Cars:** Learning to stay on the road without hitting obstacles.
- **Gaming (AlphaGo):** DeepMind's AI learned to play the complex game "Go" and defeated the world champion by playing millions of games against itself.

Key Takeaways & Tips

- **Supervised = Labeled Data:** Always look for an output column.
- **Unsupervised = Unlabeled Data:** Look for patterns, groups, or simplifications.
- **Semi-supervised = Cost-Effective:** Use it when labeling is too expensive.
- **Reinforcement = Trial & Error:** No data at the start; learn from rewards.
- **Important Repetition:** Always identify your problem type (Regression or Classification) before starting a Supervised Learning project. It changes which algorithm you will use!